

Network Traffic Analysis & DNS Protocol Inspection

TECHNICAL CASE STUDY • GOOGLE CYBERSECURITY PROFESSIONAL CERTIFICATE

1. Executive Summary

In modern Security Operations Centers (SOCs), microscopic analysis of packet captures stands as a foundational mechanism for validating protocol integrity, diagnosing underlying infrastructure anomalies, and identifying malicious command-and-control (C2) communication vectors.

This technical portfolio entry documents a rigorous protocol inspection methodology executed within a sandboxed laboratory environment. The assessment targets an encapsulated packet capture archive (`sample.pcap`) to reconstruct, trace, and dissect the transactional properties of the **Domain Name System (DNS)** application layer protocol running over the **User Datagram Protocol (UDP)**. Through granular display-filter execution and structural layer parsing, human-readable egress resolutions are cross-referenced explicitly against physical, lower-tier structural destination network targets.

2. Technical Objectives & Scope

- **Ingress & Display Filter Isolation:** Minimize ambient network chatter by isolating transactional layers specifically mapped to standard network naming services.
- **Structural Payload Decoupling:** Dissect the header properties of OSI Layer 4 (Transport) and Layer 7 (Application) architectures to map operational metrics.
- **Hexadecimal Verification:** Correlate abstract visual protocol structures within the Wireshark GUI tree layout against the raw, underlying bytecode array streams.
- **Authoritative Infrastructure Mapping:** Resolve downstream record declarations to trace precise logical endpoints mapping back to external host targets.

3. Procedural Analysis & Environmental Controls

3.1 Environment Variables

The forensic operation was deployed inside a secure, monitored environment utilizing the Wireshark Packet Analyzer engine. The underlying packet telemetry cataloged a gross population of 200 atomic capture frames.

3.2 Display Filter Ingestion

To isolate the investigation to network name resolution requests, a strict network boundary constraint filter was applied directly to the analyzer interface display pipeline:

```
udp.port == 53
```

This constraint successfully throttled active visibility down to 22 relevant packets, representing precisely 11.0% of the aggregate capture landscape, effectively neutralizing irrelevant system noise and background broadcasts.

3.3 Granular Layer Parsing (Packet No. 9)

Packet No. 9 was selected as the foundational baseline transaction, captured at relative timestamp execution metric 8.637619 seconds. Decoupling the architectural frame stack revealed the following configuration across the Open Systems Interconnection (OSI) hierarchy:

- **Network Layer (OSI Layer 3):** Handled natively via Internet Protocol Version 4 (IPv4). The packet header specifies an internal origin point source configuration of 172.21.224.2 routing toward a downstream destination address of 169.254.169.254.
- **Transport Layer (OSI Layer 4):** Executed asynchronously via User Datagram Protocol (UDP). The transaction originated out of a dynamically assigned high-order client source port (Src Port: 59398), bound directly to the standardized, authoritative well-known naming service listener interface (Dst Port: 53).
- **Application Layer (OSI Layer 7):** Initialized an active, outbound Domain Name System (query) subtree encapsulation.

4. Forensic Artifacts & Core Metrics

4.1 Dissected Structural Payload View

The table below isolates the explicit structural telemetry metadata evaluated inside the application layer subtree parsing pane during deep inspection of Packet 9:

Protocol Metric Field	Extracted Diagnostic Telemetry Value
Transaction Reference ID	0x0c26 (Unique cryptographic tracking index)
DNS Operational Code	Standard Query Flags (0x0100)
Total Request Index Counts	Questions: 1; Answer RRs: 0; Authority RRs: 0; Additional RRs: 0
Target Resolution String	opensource.google.com
Query Core Parameters	Type A (IPv4 Host Address Request), Class IN (Internet Architecture)

4.2 Hexadecimal/ASCII Byte Stream Alignment

A direct correlation assessment maps visual tree abstractions accurately against the raw bytecode layout embedded natively in the payload structure:

0000	42 01 ac 15 e0 01 42 01	ac 15 e0 02 08 00 45 00	B.....B...E.
0010	00 43 6b da 40 00 40 11	ee ba ac 15 e0 02 a9 fe	.Ck.@.@.....
0020	a9 fe e8 06 00 35 00 2f	e0 55 0c 26 01 00 00 01	5./U.&....
0030	00 00 00 00 00 00 0a 6f	70 65 6e 73 6f 75 72 63	opensour
0040	65 06 67 6f 6f 67 6c 65	03 63 6f 6d 00 00 01 00	e.google.com....
0050	01		.

Downstream Resolution Finding: Following the baseline query execution timeline forward to track subsequent authoritative response chains, the target string request identifier `opensource.google.com` aligns explicitly to the active resolved destination infrastructure host layout: **172.21.224.1**.

5. Applied Core Cybersecurity Competencies

NETWORK PROTOCOL AUDITING

Validated direct professional capability in mapping complex network conversations across asymmetric layers, tracking low-level structural wrappers down to explicit destination sockets.

TRAFFIC TELEMETRY ISOLATION

Demonstrated analytical mastery over massive diagnostic data logs, utilizing targeted boolean constraint syntax to completely exclude non-essential operational chatter, reducing analyst overhead and streamlining threat triage windows.

FORENSIC BYTE TRANSLATION

Established technical fluency in low-level telemetry structures by successfully tracing raw algorithmic data blocks back to machine-readable hex sequences and ASCII string fragments.